3.10 ANGIOGRAPHY: TOF, PC & CONTRAST-ENHANCED MRA

VISUALIZING BLOOD FLOW: NONCONTRAST OR CONTRAST—THE RIGHT TOOL FOR THE RIGHT VESSEL

MR Angiography (MRA) uses flow-related or contrast enhancement to depict arteries and veins.
Choose the technique based on vessel size, location, flow velocity, and clinical question.

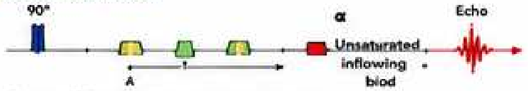
KEY CONCEPT

- TOF = time-of-flight; flow-related enhancement (inflow effect).
- PC = phase-contrast; velocity-encoded flow measurement.
- CE-MRA = gadolinium contrast for highest SNR and speed.
- Each technique has strengths, limitations, and ideal use cases.

↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

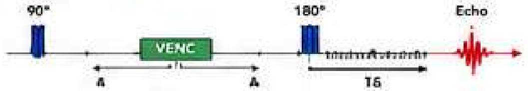
1 HOW MRA WORKS: THE BASICS

TIME-OF-FLIGHT (TOF)



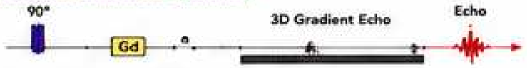
- Repeated RF excitations saturate stationary tissue.
- Inflowing blood not yet saturated → appears bright.
- Best for high-velocity, in-plane or through-plane flow.

PHASE-CONTRAST (PC)



- Flow causes phase shift proportional to velocity.
- Can measure velocity and direction (magnitude + phase).
- Best for quantification and complex flow assessment.

CONTRAST-ENHANCED (CE-MRA)



- Gadolinium shortens T1 → vessels enhance.
- Provides highest SNR and fastest coverage.
- Best for CTA alternative, large vessels, and complex anatomy.

★ All MRA techniques rely on good bolus timing, fat suppression, and optimized 3D acquisition.

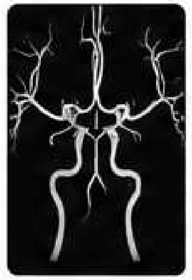
2 MRA TECHNIQUE COMPARISON

Feature	TOF MRA	PC MRA	CE-MRA
Contrast Agent	No	No	Yes (Gd)
Mechanism	Flow-related inflow effect	Velocity-encoded phase	T1 shortening (contrast)
Depicts	Arteries (best)	Arteries & Veins (flow & velocity)	Arteries & Veins (best SNR)
Best For	Intracranial, neck arteries	Flow quantification, valves, shunts	Peripheral vessels, Aorta, renal, complex anatomy
Flow Dependence	High (needs flow)	Yes (quantified)	No (contrast based)
Spatial Resolution	High	Moderate	Very High
Scan Time	Longer	Longer	Shortest
SNR	Moderate	Moderate	Highest
Velocity Sensitivity	High (inflow)	Set by VENC	Low
Artifacts	Saturation, slow flow, turbulence	Aliasing, eddy, phase wrap	Timing, venous contamination
Typical Use	Screening, follow-up	Quantitative, research, valves	Diagnostic, pre-op, emergent

ⓘ No single technique is perfect—choose based on the clinical question.

3 IMAGE APPEARANCE EXAMPLES (1.5T)

TOF MRA (Brain TOF)



- Bright arteries
- Good for intracranial circulation
- Saturation of slow flow

PC MRA (PC-VENC 80 cm/s)

- Flow direction encoded
- Veins can be visualized
- Phase scale (set by VENC)

CE-MRA (Renal Arteries)

- Bright arterial branches
- High SNR and detail
- Fast acquisition

ⓘ TOF best for brain & neck arteries; PC for flow data; CE for coverage & detail.

4 WHEN TO USE WHICH?

Clinical Scenario / Goal	Best Technique	Why?
Screening intracranial aneurysm	TOF MRA	Noninvasive, no contrast, good for Circle of Willis.
Carotid stenosis evaluation	TOF or CE-MRA	TOF for screening; CE for better stenosis quantification.
Renal artery stenosis	CE-MRA	High spatial resolution and SNR.
Aortic aneurysm / dissection	CE-MRA	Wide coverage, fast, high detail.
Pediatric / young patient	TOF or PC	Avoid contrast if possible.
Shunt (e.g., ASD, VSD, PDA)	PC-MRA	Quantify flow (Qp:Qs), direction.
Valve disease (regurgitation)	PC-MRA	Quantify flow volume and fraction.
Venous thrombosis (CTV)	CE-MRA	Best venous opacification.
Arteriovenous malformation	TOF + CE-MRA	TOF for feeders; CE for nidus/drainage.
Pre-op mapping (tumors)	CE-MRA	Detailed vascular roadmap.

5 TOF MRA: KEY POINTS

PROS

- ☑ No contrast needed
- ☑ Good for high-velocity flow
- ☑ High spatial resolution
- ☑ Excellent for intracranial arteries
- ☑ Safe for renal impairment

LIMITATIONS

- Saturation of slow or in-plane flow
- Flow direction dependent
- Longer scan times
- Susceptible to turbulence artifacts

PARAMETERS THAT MATTER

- Short TR / TE
- Flip angle: 18°–25° (optimize SNR & contrast)
- Narrow slice / 3D slab
- Flow compensation ON
- Good fat suppression

6 PC MRA: KEY POINTS

PROS

- ☑ Quantitative (velocity, flow volume)
- ☑ Measures direction & magnitude
- ☑ No contrast needed
- ☑ Excellent for complex flow (planes/valves/shunts)

LIMITATIONS

- Lower SNR than TOF/CE
- Longer scan times
- Aliasing if VENC too low
- Sensitive to patient motion

PARAMETERS THAT MATTER

- Set VENC to expected velocity (cm/s)
- Use bipolar gradients
- Thin slices; good spatial resolution
- Retrospective gating if needed
- Phase contrast direction perpendicular to flow

7 CE-MRA: KEY POINTS

PROS

- ☑ Highest SNR and contrast
- ☑ Fast acquisition
- ☑ Wide coverage
- ☑ Less flow dependent
- ☑ Best for peripheral & abdominal

LIMITATIONS

- Requires gadolinium
- Risk in severe renal failure (NSF)
- Venous contamination if timing is wrong
- Contrast timing critical

PARAMETERS THAT MATTER

- 3D T1-weighted (VIBE/THRIVE/FLASH)
- Timing: Bolus tracking or test bolus
- High spatial resolution
- Fat suppression (SPAIR/FS)
- Good bolus injection (rate 2–4 mL/s, then saline flush)

8 COMMON ARTIFACTS & FIXES

Artifact	Cause	Fix / Reduce
Saturation	Slow flow / long TR / in-plane flow	Flow comp ON, optimize flip angle, shorter TR, use CE-MRA
Flow-related signal loss	Turbulence, susceptibility	Shorter TE, flow comp, change angle
Venous contamination (CE-MRA)	Late arterial phase	Bolus tracking, earlier timing, test bolus
Gibbs ringing	High contrast edges	Increase matrix, bandwidth
Motion artifacts	Patient movement	Immobilize, faster scan, breath hold (if possible)

8 ENGINEER CHECKLIST

- ☑ Correct technique selected for clinical goal?
- ☑ Patient prepped: Screening, IV access (if CE)?
- ☑ Parameters optimized (TR, TE, FA, bandwidth)?
- ☑ Flow comp ON for TOF / PC?
- ☑ Proper VENC set for PC?
- ☑ Timing method verified for CE (bolus tracking)?
- ☑ Fat suppression effective?
- ☑ Coil elements appropriate and working?
- ☑ Review MIP / source images for artifacts?
- ☑ Compare with prior studies if available.

QUICK REFERENCE

TOF = Flow in
PC = Flow measured
CE = Contrast enhanced

TOF → High flow arteries
PC → Quantify flow (velocity, volume)
CE → Highest detail & coverage

10 CLINICAL EXAMPLES

Circle of Willis	TOF MRA	Aneurysm, stenosis, vascular variants
Carotid Arteries	TOF or CE-MRA	Stenosis, dissection
Renal Arteries	CE-MRA	Stenosis, fibromuscular dysplasia
Aortic Aneurysm	CE-MRA	Size, extent, branches
Congenital Heart (ASD)	PC-MRA	Qp:Qs, shunt flow
Peripheral Runoff	CE-MRA	PAD, pre-op planning

11 PARAMETERS THAT MATTER MOST (SUMMARY)

Parameter	TOF MRA	PC MRA	CE-MRA
TR	Short	Set by sequence	Short
TE	Short	Short	Shortest
Flip Angle	18°–25°	N/A	10°–20°
Slice / Resolution	High	Moderate–High	High (isotropic)
Flow Comp	ON	Yes (mandatory)	Usually OFF
Fat Suppression	Strongly recommended	Recommended	Strongly recommended
Timing	N/A	N/A	Bolus tracking / test bolus

12 PRACTICAL TIPS

- Always review MIP and source images.
- Use multiple planes to confirm findings.
- Adjust slab thickness and flip angle for TOF optimization.
- For PC, ensure phase wrap is corrected (adjust VENC).
- For CE, start with bolus tracking over target artery.
- Inject at rate 2–4 mL/s Gd, then 20 mL saline flush.
- Document and communicate any limitations.

REMEMBER

Right technique.
Right timing.
Right parameters.
= Right answer.

KEY TAKEAWAY

TOF = flow in and go.
PC = flow measured.
CE = see it all.